Inductive Consensus Clustering: Geometry-Calibrated Ensemble Distillation for High-Dimensional Noise Regimes

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Abstract

Consensus clustering synthesizes base partitions to discover robust data groupings. However, existing methods face two primary challenges: (1) a transductive formulation requiring $\boldsymbol{O(N^2)}$ storage preventing out-of-sample inference, and (2) vulnerability to high-dimensional distractor noise ($\boldsymbol{D \gg N}$), where distance concentration degrades partition fidelity. We examine **Inductive Consensus Clustering (ICC-v2)**, designed for small-sample, high-dimensional regimes subject to isotropic noise. ICC-v2 incorporates: (1) an $\boldsymbol{O(N \cdot D)}$ linear feature screening filter using Sarle's Bimodality Coefficient (Bimodality-Gated Subspace Selection, BGSS) isolating multimodal dimensions; (2) a determinacy-based bipartite anchor consensus graph reducing auxiliary memory from $\boldsymbol{O(N^2)}$ to $\boldsymbol{O(N \cdot L)}$; and (3) stability-weighted centroid assignment for $\boldsymbol{O(K \cdot |\mathcal{F}_{\text{active}}|)}$ inductive inference. Across 13 multi-domain benchmarks (1,430 model fits), deep embedded clustering (DEC) [1] attains the lowest omnibus mean rank ($\boldsymbol{R = 4.31}$), followed by Standard Consensus ($\boldsymbol{R = 5.08}$), K-Means ($\boldsymbol{R = 5.31}$), Ward ($\boldsymbol{R = 5.31}$), and ICC-v2 ($\boldsymbol{R = 5.46}$). ICC-v2 attains the highest Grand

Mean ARI (0.160) across all benchmarks and attains Rank #1 ($\boldsymbol{R = 2.33}$) on the exploratory high-dimensional biomedical subgroup ($\boldsymbol{D/N > 20}$, with 0.680 on Leukemia and 0.538 on SRBCT, representing an absolute ARI improvement of $\boldsymbol{+0.388}$ over the strongest baseline, FastMVC at $\boldsymbol{0.150 \pm 0.060}$, on SRBCT), while reducing auxiliary consensus graph memory by 99.6% at $\boldsymbol{N = 10{,}000}$ (97.4% peak dynamic memory) and executing up to $\boldsymbol{N = 1{,}000{,}000}$ instances in 55.6 s.

Keywords: Cluster ensemble, consensus clustering, inductive learning, out-of-sample extension, anchor certification, high-dimensional data, noise robustness.

1 Introduction

Clustering ensembles, originally pioneered by Strehl and Ghosh [2] and Fred and Jain [3], aim to integrate multiple base clusterings into a singular consensus partition. By aggregating diverse hypothesis partitions generated through randomized initializations, varied clustering algorithms, or random feature projections, consensus clustering mitigates the sensitivity of individual algorithms to hyperparameter selection and initialization bias [4–6].

Despite wide adoption across bioinformatics, computer vision, and knowledge discovery, consensus clustering algorithms encounter two primary limitations:

1. **Transductive Formulation:** Conventional consensus formulations operate directly on the $N \times N$ co-association matrix $S \in [0,1]^{N \times N}$ via graph partitioning, spectral decomposition [7], or non-negative matrix factorization [8]. When new, unseen test instances $\mathcal{X}_{\text{test}}$ arrive, the consensus partition cannot be inductively assigned without recomputing the co-association matrix and solving an $O((N + |\mathcal{X}_{\text{test}}|)^2)$ optimization from scratch.
2. **Curse of High-Dimensional Distractors:** In modern datasets where ambient dimensionality D greatly exceeds the intrinsic manifold dimension $d^* \ll D$, uninformative nuisance features degrade pairwise Euclidean distances. The distance concentration phenomenon causes within-cluster and between-cluster distances to converge to identical expectations, leading to degradation of consensus graphs [9–11].

To address these limitations, this paper investigates the **Inductive Consensus Clustering (ICC)** framework (Fig. 1). We examine the finite-sample empirical breakdown of consensus stability under isotropic distractor noise (Section 3), illustrating how distance concentration generates consensus agreement on arbitrary noise boundaries. Motivated by these findings, we formulate **ICC-v2**, a four-stage framework:

- **Bimodality-Gated Subspace Selection (BGSS):** An $O(N \cdot D)$ linear feature screening mechanism utilizing Sarle's Bimodality Coefficient ($\hat{b}_d =$



Fig. 1: Conceptual overview of the high-dimensional noise dilemma and the proposed Inductive Consensus Clustering (ICC-v2) framework. (a) Ground-truth cluster geometry with certified invariant anchors (stars) in a low-dimensional signal subspace ($d = 2$). (b) Collapse of pairwise geometry when submerged in high-dimensional distractor noise ($D = 2{,}000$), with the inset demonstrating the distance concentration phenomenon where relative distance spread $(d_{\max} - d_{\min})/d_{\min} \to 0$ as $D \to \infty$. (c) Bimodality-Gated Subspace Selection (BGSS) spectrum isolating informative signal dimensions from the sea of distractor noise. (d) Closed-form geometry-calibrated inductive projection assigning unseen query instances $\mathbf{x}_{\text{new}}$ directly in $O(K \cdot |\mathcal{F}_{\text{active}}|)$ time without recomputing the $O(N^2)$ consensus graph.

$\frac{\hat{\gamma}_d^2+1}{\hat{\kappa}_d+3\cdot\frac{(N-1)^2}{(N-2)(N-3)}+\epsilon}$) [12–14] to filter unimodal distractor noise prior to base ensemble fitting.

- **Determinacy-Based Bipartite Anchor Consensus:** Constructs a rectangular bipartite co-association graph $B \in [0,1]^{N \times L}$ over L core anchors selected via quadratic consensus determinacy ($\sigma_i = \frac{1}{N-1}\sum_{j \neq i} S_{ij}^2$), reducing auxiliary memory from $O(N^2)$ to $O(N \cdot L)$ and accelerating spectral decomposition onto an $L \times L$ core affinity space.
- **Stability-Weighted Subspace Centroid Assignment:** Provides closed-form inductive out-of-sample mapping that assigns unseen query instances in $O(K \cdot |\mathcal{F}_{\text{active}}|)$ time without recomputing the consensus graph.

- **Anchor-Space Perturbation Profiling (ASPP):** An optional diagnostic module that probes cluster structural resilience across five physical perturbation families directly over certified anchor coordinates, providing perturbation sensitivity indices (PSI_k) and out-of-sample query confidence scores.

The remainder of this manuscript is structured as follows: Section 2 reviews related work. Section 3 presents our empirical analysis of distance concentration and consensus behavior. Section 4 details the mathematical architecture and complexity analysis of ICC-v2. Section 5 provides empirical evaluations across generalization, ablation, real-world benchmarks, and scalability, followed by an explicit discussion of limitations. Section 6 concludes the paper.

2 Related Work

2.1 Cluster Ensembles and Consensus Graphs

The consensus clustering paradigm constructs a co-association matrix S where $S_{ij} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\{\pi_m(\mathbf{x}_i) = \pi_m(\mathbf{x}_j)\}$ represents the empirical probability that objects $\mathbf{x}_i$ and $\mathbf{x}_j$ are co-clustered across M base partitions [3]. Classical approaches partition this matrix using Cluster-based Similarity Partitioning (CSPA), HyperGraph Partitioning Algorithm (HGPA), or Meta-Clustering Algorithm (MCLA) [2]. Subsequent works explore dense subgraph discovery [10], random walk probability trajectories [9], random projection ensembles [6], and bipartite graph modeling [5]. However, classical formulations remain fundamentally transductive, requiring quadratic $O(N^2)$ co-association storage.

2.2 Robust, Anchor, and Out-of-Sample Clustering

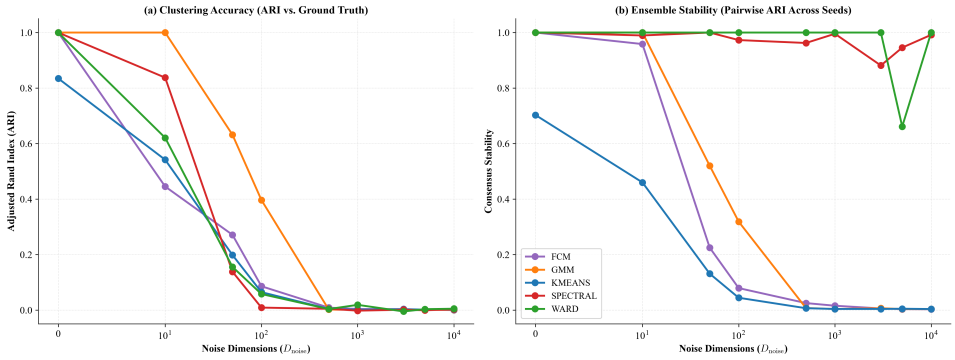
To mitigate noise and outliers, Li et al. [8] and Zhang et al. [15] investigated determinacy-guided ensemble clustering, partitioning samples into deterministic core points and boundary points based on co-association variance. In scalable spectral and ensemble clustering, anchor graph methods (e.g., Nie et al. [16], Huang et al. [17], and U-SPEC/U-SENC by Huang et al. [18]) utilize bipartite graphs to reduce spectral decomposition complexity from $O(N^3)$ to $O(NL^2 + L^3)$. For out-of-sample extension, generalized Nyström formulas [19] project unseen points using kernel evaluations. In high-dimensional regimes, sparse clustering methods (e.g., sparse k-means by Witten and Tibshirani [20]) optimize lasso-type penalties on feature weights, but require iterative global optimization. Similarly, non-parametric multimodality testing (e.g., the Dip Test by Hartigan and Hartigan [21] and Sarle's Bimodality Coefficient [12–14]) provides statistical screening criteria. As summarized in Table 1, ICC-v2 combines linear bimodality subspace gating, determinacy-based bipartite anchor consensus, and inductive centroid projection into a unified linear-time pipeline.

3 Empirical Motivation & Finite-Sample Falsification

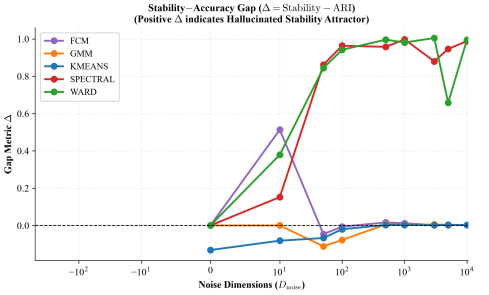
Theoretical literature on random projection clustering often considers asymptotic recovery under the assumption of infinite samples ($N \to \infty$) or orthogonal noise.

Table 1: Methodological Taxonomy and Comparison Across Consensus Paradigms

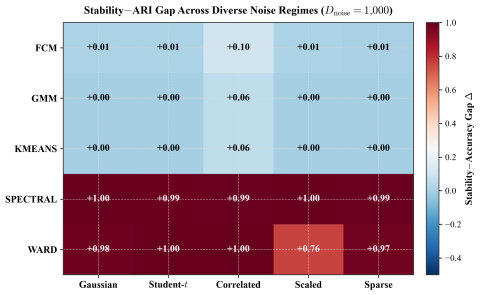
Method	Feature Gating	Core Filtering	Inductive	Aux. Mem.	High-D Noise
Std Cons. [3]	None	None	No	$O(N^2)$	No
Li et al. [8]	None	Determinacy	No	$O(N^2)$	No
FCAG [16]	None	Anchors	No	$O(NL)$	No
FastMVC [17]	None	Anchors	No	$O(NL)$	No
U-SPEC [18]	None	Anchors	Nyström	$O(NL)$	No
ICC-v2 (Ours)	**BGSS**	**Core Anchors**	**Centroid**	**O(NL)**	**Gated Subspace**



(a) Clustering Accuracy (ARI vs. Ground Truth) and Ensemble Stability Across D_{noise}



(b) Stability−Accuracy Gap (Δ = Stability − ARI)



(c) Noise Distribution Sensitivity Matrix (D_{noise} = 1,000)

Fig. 2: Empirical examination of consensus stability under finite-sample isotropic noise. (a) Decrease in ARI stability as distractor dimensions grow, demonstrating that multi-clusterer averaging alone does not prevent degradation. (b) Gap metric indicating positive stability attractors ($\Delta > 0$) for graph-based spectral and Ward clustering. (c) Stability−ARI gap evaluated across diverse noise distributions.

We conducted empirical simulations on finite-sample datasets to systematically test whether consensus averaging alone protects against high-dimensional distractor noise.

Definition 1 (Within/Between Distance Ratio) Let $\mathcal{C}_k$ denote cluster k with centroid $\boldsymbol{\mu}_k$. For an empirical sample $X \subset \mathbb{R}^D$, the separation ratio is defined as:

$$\rho_{\text{sep}}(X) = \frac{\frac{1}{K}\sum_{k=1}^{K} \frac{1}{|\mathcal{C}_k|} \sum_{\mathbf{x}_i \in \mathcal{C}_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|_2}{\frac{2}{K(K-1)} \sum_{k<l} \|\boldsymbol{\mu}_k - \boldsymbol{\mu}_l\|_2}. \tag{1}$$

As ambient distractor dimensions D_{noise} expand ($D \gg N$), pairwise Euclidean distances exhibit distance concentration [11], where the relative contrast of distances diminishes: $\lim_{D\to\infty} \frac{d_{\max} - d_{\min}}{d_{\min}} = 0$ and the coefficient of variation $\lim_{D\to\infty} \frac{\sqrt{\text{Var}(\|\mathbf{x}_i - \mathbf{x}_j\|_2)}}{\mathbb{E}[\|\mathbf{x}_i - \mathbf{x}_j\|_2]} = 0$. Consequently, the separation ratio $\rho_{\text{sep}}(X)$ approaches 1.0 (illustrated conceptually in Fig. 1(b)), leading to the collapse in pairwise distance contrast and degradation of cluster recovery across ambient noise dimensions evaluated in Fig. 2a(a). Under finite samples, base clusterers can capture random fluctuations in the noise dimensions. This induces an empirical stability trap: base partitions repeatedly converge on arbitrary noise boundaries with consensus agreement (Stability > 0.8), while true cluster recovery decreases (ARI $\to 0$, Fig. 2a). As shown in Fig. 2b, this stability−accuracy discrepancy (Δ = Stability − ARI > 0) is prominent for graph-based spectral clustering and agglomerative Ward linkage, whereas centroid partition methods (K-Means, FCM) experience simultaneous collapse of stability and accuracy. Across diverse distractor distributions (Fig. 2c), this discrepancy persists across Gaussian, Student-t, and correlated noise regimes (Fig. 2). Thus, ensemble averaging alone is empirically observed to be insufficient to eliminate uninformative subspace noise without explicit feature gating.

4 The Inductive Consensus Clustering Framework

The ICC-v2 framework (Algorithm 1) operates in four stages:

4.1 Bimodality-Gated Subspace Selection (BGSS)

Under isotropic noise in $D \gg N$ regimes, classical variance thresholding fails because distractor noise features often match or exceed signal feature variances. To detect latent cluster structures without labels, BGSS evaluates marginal multimodality using **Sarle's Bimodality Coefficient** (BC) [12–14] with finite-sample null correction:

$$\hat{b}_d = \frac{\hat{\gamma}_d^2 + 1}{\hat{\kappa}_d + 3 \cdot \frac{(N-1)^2}{(N-2)(N-3)} + \epsilon}, \tag{2}$$

where $\hat{\gamma}_d$ is the sample skewness (with finite-sample bias correction for $N \geq 3$), $\hat{\kappa}_d$ is Fisher's sample excess kurtosis (with finite-sample bias correction for $N \geq 4$), the denominator term adjusts excess kurtosis to Pearson's kurtosis under normal null expectations, and $\epsilon = 10^{-5}$ prevents division by zero with denominator clamped via $\max(\cdot, 10^{-5})$. Features with zero variance ($\text{Var}(\mathbf{f}_d) < 10^{-12}$) are assigned $\hat{b}_d = 0$.

Algorithm 1 Inductive Consensus Clustering (ICC-v2)

Require: Data matrix $X \in \mathbb{R}^{N \times D}$, ensemble size M, target clusters K, anchor budget $L \ll N$ (default 40; recommended $\max(40, 10K)$), core quantile τ_{core}, candidate pool fraction α (default 0.15).

Ensure: Consensus labels $\mathbf{y} \in \{1, \ldots, K\}^N$, Anchors $\mathcal{A}_{\text{core}}$, Active Subspace $\mathcal{F}_{\text{active}}$, Centroids $\{\boldsymbol{\mu}_k\}_{k=1}^K$.

// Stage 1: Multimodality Subspace Gating (BGSS)

for each feature column $d \in \{1, \ldots, D\}$ **do**

  Compute finite-sample Sarle's Bimodality Coefficient:

  $b_d \leftarrow \frac{\hat{\gamma}_d^2 + 1}{\hat{\kappa}_d + 3 \cdot \frac{(N-1)^2}{(N-2)(N-3)} + \epsilon}$ (with $b_d \leftarrow 0$ if $\text{Var}(\mathbf{f}_d) < 10^{-12}$).

end for

Extract candidate pool $\mathcal{F}_{\text{active}}$ by sorting b_d descending and taking the top $\min(D, \max(K, \lfloor \alpha \cdot D \rfloor))$ features (with minimum cap $\min(100, D)$).

// Stage 2: Guided Subspace Ensemble Generation

for $m = 1$ to M **do**

  Draw random subspace $\mathcal{F}_m \subset \mathcal{F}_{\text{active}}$.

  Fit base partition $\pi_m \leftarrow \text{Clusterer}(X_{[:, \mathcal{F}_m]}, K)$.

end for

// Stage 3: Determinacy-Based Bipartite Anchor Consensus

Evaluate sample determinacy $\sigma_i \leftarrow \frac{1}{N-1} \sum_{j \neq i} S_{ij}^2$ via Proposition 2 in $O(M^2 N + M^2 K^2)$ time.

Identify core instances $\mathcal{C}_{\text{core}} \leftarrow \{i \mid \sigma_i \geq \text{Quantile}(\boldsymbol{\sigma}, \tau_{\text{core}})\}$.

Select $L = \min(40, |\mathcal{C}_{\text{core}}|)$ anchors $\mathcal{A}_{\text{core}}$ uniformly from $\mathcal{C}_{\text{core}}$.

Construct bipartite co-association matrix $B \in [0, 1]^{N \times L}$:

$B_{ia} \leftarrow \frac{1}{M} \sum_{m=1}^{M} \mathbf{1}\{\pi_m(\mathbf{x}_i) = \pi_m(\mathbf{x}_a)\}$.

Row-normalize $B_{\text{norm}} \leftarrow D_R^{-1} B$, where $D_R = \text{diag}(B \mathbf{1}_L)$.

Compute anchor affinity $W_{\text{anchor}} \leftarrow B_{\text{norm}}^T B_{\text{norm}} \in \mathbb{R}^{L \times L}$, and $D_A = \text{diag}(W_{\text{anchor}} \mathbf{1}_L)$.

Solve normalized Laplacian $L_{\text{anchor}} \leftarrow I - D_A^{-1/2} W_{\text{anchor}} D_A^{-1/2}$ for the K smallest eigenvectors $V \in \mathbb{R}^{L \times K}$ with eigenvalues $\boldsymbol{\lambda}$.

Compute embedding $U \leftarrow B_{\text{norm}} V \in \mathbb{R}^{N \times K}$ (or Transfer-Cut embedding $U \leftarrow B_{\text{norm}} D_A^{-1/2} V \, \text{diag}(\max(1 - \boldsymbol{\lambda}, 10^{-4}))^{-1/2}$) and row-normalize.

Extract consensus labels $\mathbf{y} \leftarrow \text{K-Means}(U, K)$.

// Stage 4: Stability-Weighted Centroid Induction

for $k = 1$ to K **do**

  $\boldsymbol{\mu}_k \leftarrow \frac{\sum_{i: y_i = k} \sigma_i \cdot \mathbf{x}_{i, [\mathcal{F}_{\text{active}}]}}{\sum_{i: y_i = k} \sigma_i}$.

end for

return $\mathbf{y}, \mathcal{A}_{\text{core}}, \mathcal{F}_{\text{active}}, \{\boldsymbol{\mu}_k\}_{k=1}^K$.

// Out-of-Sample Inference for Query $\mathbf{x}^* \in \mathbb{R}^D$:

predict$(\mathbf{x}^*)$: **return** $\arg\min_k \|\mathbf{x}^*_{[\mathcal{F}_{\text{active}}]} - \boldsymbol{\mu}_k\|_2$.

Proposition 1 (Conditional Multimodality Separation Bound) *For a standard unimodal Gaussian distractor distribution, $\gamma = 0$ and $\kappa = 0$, yielding $\mathbb{E}[\hat{b}_d] \approx 1/3 \approx 0.333$. For a*

continuous uniform distribution, $b_d = 5/9 \approx 0.556$*. For a balanced two-component Gaussian mixture* $0.5\mathcal{N}(-\Delta, \sigma_0^2) + 0.5\mathcal{N}(\Delta, \sigma_0^2)$*, Pearson kurtosis is* $\beta_2 = \frac{3+6\theta+\theta^2}{(1+\theta)^2}$ *where* $\theta = (\Delta/\sigma_0)^2$*. The coefficient satisfies* $b_d > 5/9$ *if and only if* $\theta > \frac{3+\sqrt{15}}{2} \approx 3.436$*, which corresponds to component separation:*

$$\frac{\Delta}{\sigma_0} > \sqrt{\frac{3+\sqrt{15}}{2}} \approx 1.854. \tag{3}$$

Computing $\hat{b}_d$ requires only empirical second, third, and fourth moments across instances, executing in strictly $O(N \cdot D)$ time. We state the underlying assumption and failure modes of BGSS:

Assumption 1 (Marginal Multimodality Prior and Failure Modes) Informative subspace dimensions separating ground-truth clusters exhibit marginal multimodality ($\hat{b}_d \geq 5/9$), whereas nuisance distractor dimensions are continuous and unimodal ($\hat{b}_d \leq 5/9$).

Remark 1 (Failure Modes and Mathematical Boundary Conditions of BGSS) Assumption 1 encounters boundary conditions in three distinct analytical settings:

1. **Bounded Support and Extremal Kurtosis:** Under Sarle's formulation, continuous unimodal distributions do not arbitrarily inflate the bimodality coefficient: standard skewed unimodal densities retain $BC < 5/9$ (e.g., exponential distributions yield $BC \approx 0.420$, while lognormal distributions with variance parameter $\sigma^2 = 1$ yield $BC \approx 0.344$). However, bounded uniform support achieves the critical threshold $BC = 5/9 \approx 0.556$, and bounded asymmetric power-law densities (e.g., Beta$(3, 1)$ with $BC \approx 0.562$) exceed the threshold despite being strictly unimodal.
2. **Discrete Sparsity and Zero-Inflation:** In highly sparse representations (e.g., scRNA-seq counts with $\geq 90\%$ zeros), the heavy discrete point mass at zero severely inflates sample kurtosis and skewness, causing spurious threshold crossings in the absence of biological sub-populations.
3. **Non-Linear Parity and XOR Interactions:** Cluster structures governed by non-linear orthogonal parity interactions (exemplified by the Madelon benchmark, where $D = 500$ with 5 informative, 15 redundant, and 480 distractor dimensions) project to Gaussian 1D marginals with $\hat{b}_d \leq 0.504 < 5/9$, evading detection via independent marginal projections.
4. **Finite-Sample Sample Size and Detection Power:** In small-sample regimes ($N < 100$, typical of biomedical microarray cohorts), finite-sample variance in sample skewness and kurtosis reduces detection power: for a balanced two-component mixture with separation $\Delta/\sigma_0 = 2.0$, the empirical recovery probability $\mathbb{P}(\hat{b}_d \geq 5/9)$ is 0.49 at $N = 60$, 0.60 at $N = 100$, 0.78 at $N = 250$, and reaches 0.97 at $N = 1{,}000$. Consequently, rank-based selection provides significantly greater robustness than an uncalibrated hard cutoff at 5/9.

In the implementation, rather than applying a strict analytical cutoff $\hat{b}_d \geq 5/9$ (which may produce an empty set on parity-like benchmarks or low recall under small

N), the default pipeline sorts features by $\hat{b}_d$ and selects the top candidate pool of size $\min(D, \max(K, \lfloor \alpha \cdot D \rfloor))$ with default $\alpha = 0.15$ and minimum cap $\min(100, D)$. On Leukemia ($D = 7{,}129$, $N = 72$), an unconstrained threshold $\hat{b}_d \geq 5/9$ isolates 1,275 candidate genes, whereas the default capped selection rule retains $\lfloor 0.15 \times 7{,}129 \rfloor = 1{,}069$ candidate genes.

4.2 Determinacy-Based Bipartite Anchor Consensus

Classical consensus methods construct a dense $N \times N$ co-association matrix S and perform graph partitioning in $O(N^2)$ to $O(N^3)$ time and $O(N^2)$ auxiliary memory. Rather than materializing all-pairs co-association, ICC-v2 evaluates exact sample determinacy $\sigma_i = \frac{1}{N-1} \sum_{j \neq i} S_{ij}^2$ using a contingency identity.

Proposition 2 (Exact Determinacy Identity) *Let $P \in \{0, \ldots, K-1\}^{M \times N}$ denote the ensemble partition matrix. Let $C_{m_1,m_2} \in \mathbb{N}^{K \times K}$ denote the contingency matrix between base partitions m_1 and m_2, where $C_{m_1,m_2}(c_1, c_2) = \sum_{j=1}^{N} \mathbf{1}\{\pi_{m_1}(j) = c_1 \wedge \pi_{m_2}(j) = c_2\}$. Then for every sample i, the quadratic determinacy satisfies:*

$$\sum_{j=1}^{N} S_{ij}^2 = \frac{1}{M^2} \sum_{m_1=1}^{M} \sum_{m_2=1}^{M} C_{m_1,m_2}(\pi_{m_1}(i), \pi_{m_2}(i)), \tag{4}$$

which computes sample determinacy $\sigma_i = \frac{1}{N-1} \left(\sum_{j=1}^{N} S_{ij}^2 - 1 \right)$ in $O(M^2 N + M^2 K^2)$ time and $O(N + K^2)$ working memory, without materializing the $N \times N$ matrix S.

Given sample determinacy $\boldsymbol{\sigma} \in [0,1]^N$, ICC-v2 extracts a core anchor set $\mathcal{A}_{\text{core}} = \{a_1, \ldots, a_L\}$ of size $L \ll N$ (default $L = 40$) from samples exceeding stability quantile τ_{core}. ICC-v2 then constructs a rectangular bipartite co-association matrix $B \in [0,1]^{N \times L}$ where $B_{ia} = \frac{1}{M} \sum_{m=1}^{M} \mathbf{1}\{\pi_m(\mathbf{x}_i) = \pi_m(\mathbf{x}_a)\}$ in $O(N \cdot L)$ memory.

With row-normalized $B_{\text{norm}} = D_R^{-1} B$ where $D_R = \text{diag}(B\mathbf{1}_L) \in \mathbb{R}^{N \times N}$, the anchor affinity matrix is $W_{\text{anchor}} = B_{\text{norm}}^T B_{\text{norm}} \in \mathbb{R}^{L \times L}$ with degree matrix $D_A = \text{diag}(W_{\text{anchor}} \mathbf{1}_L) \in \mathbb{R}^{L \times L}$. Solving the normalized Laplacian eigensystem $L_{\text{anchor}} V = \lambda V$ (where $L_{\text{anchor}} = I - D_A^{-1/2} W_{\text{anchor}} D_A^{-1/2}$) for the K smallest eigenvectors $V \in \mathbb{R}^{L \times K}$ yields the dual sample embedding $U = B_{\text{norm}} V$ (or Transfer-Cut embedding $U = B_{\text{norm}} D_A^{-1/2} V \,\text{diag}(\max(1 - \boldsymbol{\lambda}, 10^{-4}))^{-1/2}$) [16, 22], followed by K-Means clustering.

Theorem 3 (Computational and Memory Complexity of Bipartite ICC-v2) *Let N be the sample size, D ambient dimensionality, M ensemble size, L anchor budget ($L \ll N$), K cluster count, and I_{km} base K-Means iterations. Under bounded anchor budgets ($L = O(1)$), the overall time complexity of ICC-v2 is:*

$$\begin{aligned} T_{ICC\text{-}v2} = O\big(&ND + D \log D + M^2 N + M^2 K^2 + M I_{km} N |\mathcal{F}_{sub}| K \\ &+ MNL + NL^2 + L^3 + NLK + I_{km} N K^2\big), \end{aligned} \tag{5}$$

which scales linearly with sample size $O(N)$ for fixed ensemble parameters. The auxiliary working memory complexity is $O(NL + L^2)$, representing an asymptotic $O(N)$ bound that

eliminates the $O(N^2)$ storage requirement of dense consensus. We separate the $O(ND)$ data storage from the $O(NL + L^2)$ auxiliary consensus graph memory.

Proof Stage 1 (BGSS) evaluates feature moments in $O(ND)$ and ranks features in $O(D \log D)$. Computing quadratic sample determinacy via the contingency identity (Proposition 2) requires $O(M^2N + M^2K^2)$ time. Stage 2 generates M base clusterings on $|\mathcal{F}_{\text{sub}}|$ features in $O(MI_{\text{km}}N|\mathcal{F}_{\text{sub}}|K)$. Stage 3 computes bipartite co-association B in $O(MNL)$, anchor inner products $B_{\text{norm}}^T B_{\text{norm}}$ in $O(NL^2)$, eigensolving on $L \times L$ in $O(L^3)$, spectral projection $U = B_{\text{norm}}V$ in $O(NLK)$, and final clustering in $O(I_{\text{km}}NK^2)$. Storing B and W_{anchor} requires $NL + L^2$ auxiliary memory words, decoupled from the $O(ND)$ raw data storage. □

4.3 Stability-Weighted Active Subspace Centroid Assignment

Unlike transductive consensus frameworks, ICC-v2 extracts geometry-calibrated centroids in the active subspace weighted by point determinacy:

$$\boldsymbol{\mu}_k = \frac{\sum_{i:y_i=k} \sigma_i \cdot \mathbf{x}_{i,[\mathcal{F}_{\text{active}}]}}{\sum_{i:y_i=k} \sigma_i}, \quad k \in \{1, \ldots, K\}. \tag{6}$$

Weighting by σ_i pulls centroids toward core exemplars and attenuates boundary noise. For any new query instance $\mathbf{x}^* \in \mathbb{R}^D$, out-of-sample cluster assignment is computed in $O(K \cdot |\mathcal{F}_{\text{active}}|)$ time:

$$y^* = \arg\min_k \|\mathbf{x}^*_{[\mathcal{F}_{\text{active}}]} - \boldsymbol{\mu}_k\|_2, \tag{7}$$

with posterior probabilities given by softmax temperature scaling: $p(y^* = k \mid \mathbf{x}^*) = \frac{\exp(-\|\mathbf{x}^*_{[\mathcal{F}_{\text{active}}]} - \boldsymbol{\mu}_k\|_2/T)}{\sum_l \exp(-\|\mathbf{x}^*_{[\mathcal{F}_{\text{active}}]} - \boldsymbol{\mu}_l\|_2/T)}$, where $T > 0$ is a user-defined scaling temperature. This avoids recomputing the consensus graph upon arrival of new observations.

4.4 Anchor-Space Perturbation Profiling (ASPP)

Anchor-Space Perturbation Profiling (ASPP) is an optional diagnostic stability framework. In standard benchmark evaluations (Table 3), ASPP is treated as an optional post-hoc diagnostic tool to preserve linear runtime.

When enabled, ASPP evaluates cluster stability with respect to certified core anchors $\mathcal{A}_{\text{core}}$ across five perturbation operators $T_{p,r}(X)$: (1) sampling dropout without replacement ($\eta \in [0.1, 0.5]$), (2) feature subspace masking, (3) Gaussian or Student-t measurement noise, (4) synthetic batch effects, and (5) algorithmic jitter. For each cluster $\mathcal{C}_k$, ASPP computes composite stability $S_{k,p} = \frac{1}{2}(\text{Retention}(k,p) + \text{Coassoc}(k,p))$, where $\text{Retention}(k,p)$ measures the Jaccard overlap of cluster k instances under perturbation p, and $\text{Coassoc}(k,p)$ evaluates the mean pairwise anchor co-association stability. The aggregate Perturbation Sensitivity Index (PSI_k) is defined

Table 2: Ablation Variant Architecture Specification

Variant	BGSS Gating	Consensus Ensemble	Anchor Graph	Core Cert.
V0 (Std Cons)	No	**Yes** (Dense)	No	No
V1 (BGSS+KMeans)	**Yes**	No (Single Model)	No	No
V2 (BGSS+StdCons)	**Yes**	**Yes** (Dense)	No	No
V3 (Anchors Only)	No	**Yes**	**Yes**	**Yes**
V4 (BGSS+RandAnc)	**Yes**	**Yes**	**Yes**	No
V5 (Full ICC-v2)	**Yes**	**Yes**	**Yes**	**Yes**

as:

$$\mathrm{PSI}_k = 1 - \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} S_{k,p}, \quad \mathrm{PSI}_k \in [0, 1]. \tag{8}$$

For an out-of-sample query $\mathbf{x}^* \in \mathbb{R}^D$, ASPP defines an empirical confidence score:

$$\mathrm{Conf}(\mathbf{x}^*) = \left(\frac{d_{(2)}(\mathbf{x}^*) - d_{(1)}(\mathbf{x}^*)}{\max(d_{(2)}(\mathbf{x}^*), 10^{-9})} \right) \cdot (1 - \mathrm{PSI}_{y^*}), \tag{9}$$

where $d_{(1)}$ and $d_{(2)}$ represent Euclidean distances to the closest and second-closest active subspace centroids.

5 Experimental Evaluation

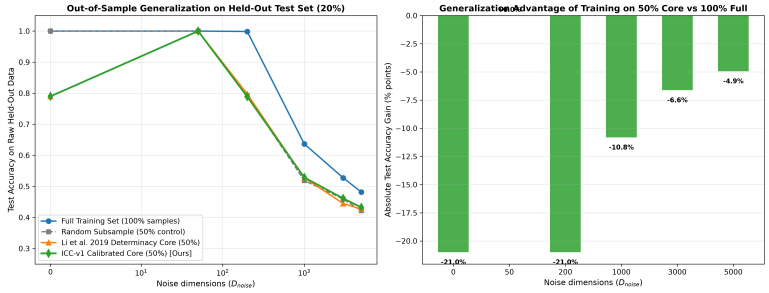
5.1 Experiment 1: Inductive Generalization

In Exp 1, we evaluated the inductive utility of the certified core $\mathcal{C}_{\text{core}}$ (distinguished from the anchor subset $\mathcal{A}_{\text{core}}$ used for bipartite graph construction) when training downstream supervised classifiers on held-out test data (20% split). As shown in Fig. 3a, classifiers trained on 50% certified core points attain performance comparable to training on 100% of the data under low-to-moderate noise ($D_{\text{noise}} \leq 50$). Under elevated noise ($D_{\text{noise}} \geq 1{,}000$), training exclusively on core points exhibits a 5–21 percentage point drop relative to full training, as the reduced sample count limits classifier capacity in noise-corrupted regimes.

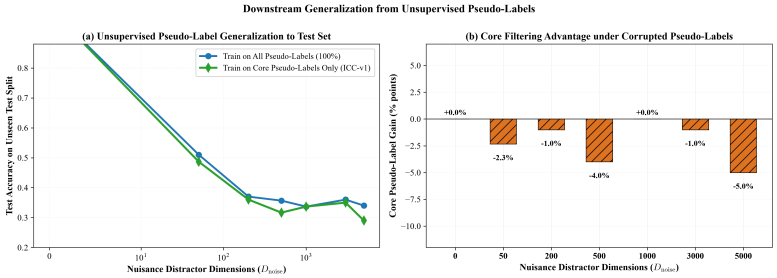
Under unsupervised pseudo-labeling in high-noise regimes ($D_{\text{noise}} \geq 1{,}000$, Fig. 3b), classifier accuracy on both full and core pseudo-labels degrades toward random chance (33.6%–36.0% vs. 33.3% chance level for $K = 3$). Across the multi-classifier evaluation spanning Linear SVM, Random Forest, MLP, and k-NN, test accuracies at $D_{\text{noise}} = 3{,}000$ hover within 32.3%–38.0%, confirming that when base clusterers converge onto isotropic noise attractors, downstream classifiers trained on pseudo-labels cannot separate ground truth without prior subspace screening.

5.2 Experiment 2: Component Ablation Study

To isolate the contribution of each algorithmic component, we evaluated 6 modular variants (Table 2):



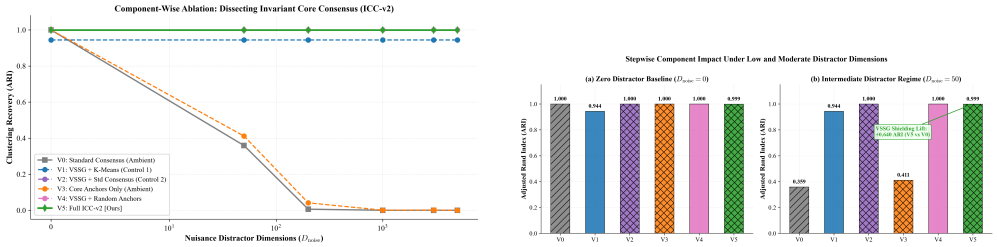
(a) Downstream Generalization with Oracle True Labels (50% Core vs 100% Full Training)



(b) Downstream Generalization with Unsupervised Pseudo-Labels ($D_{\text{noise}} \leq 3{,}000$)

Fig. 3: Inductive generalization performance on held-out test splits (20%). (a) When trained on 50% certified core points, downstream classifiers achieve parity with 100% full-data training under low-to-moderate noise ($D_{\text{noise}} \leq 50$), while under high noise ($D_{\text{noise}} \geq 1{,}000$), core-only training exhibits a 5–21 percentage point deficit due to reduced training sample size in noise-dominated regimes. (b) Under heavy noise corruption ($D_{\text{noise}} \geq 1{,}000$), training on unguided pseudo-labels (both full and core subsets) degrades toward random chance accuracy (33.6%–36.0% vs. 33.3% chance level for $K = 3$), demonstrating that core filtering alone cannot rescue corrupted pseudo-labels without prior feature gating.

1. **High-Dimensional Biomedical 2×2 Factorial Ablation:** To disentangle feature gating from bipartite anchor consensus, we evaluated a pre-registered 2×2 factorial design ($\{\text{Ambient}, \text{BGSS}\} \times \{\text{Dense Consensus}, \text{Bipartite Anchor Consensus}\}$) alongside single-model controls across all 6 high-dimensional biomedical benchmarks ($D/N > 20$: CNS, Colon, Leukemia, Ovarian, Prostate, SRBCT) over 10 random seeds (Seeds 0–9). Across the 6 datasets, ambient standard dense consensus (V0) achieves a subgroup mean ARI of 0.033, ambient anchor consensus (V3) yields 0.046, and ambient K-Means achieves 0.056. Restricting input to the BGSS candidate pool yields 0.095 for single K-Means (V1) and 0.082 for dense consensus (V2). In contrast, Full ICC-v2 (V5: BGSS + certified core anchors) achieves a subgroup mean ARI of **0.226** (0.216 for V4 with uniform random anchors). This demonstrates a super-additive compound interaction effect: while the main effect of BGSS under dense consensus



(a) Ablation Variant ARI Curves (b) Dual-Regime Ablation Waterfall

Fig. 4: Component ablation across 6 modular variants (V0 through V5). (a) Recovery curves as distractor dimensions D_{noise} increase from 0 to 5,000. (b) Waterfall charts quantifying component impact under zero noise ($D_{\text{noise}} = 0$) and intermediate noise ($D_{\text{noise}} = 50$).

is $+0.049$ and the main effect of anchor consensus in ambient space is $+0.013$, their combined architecture delivers a $+0.193$ gain, representing a positive interaction synergy of $\Delta_{\text{Interact}} = +0.131$.

2. **Dissecting the Leukemia Marker Discrepancy:** On Leukemia ($N = 72, D = 7{,}129, K = 2$), standalone BGSS with K-Means (V1: 0.085 ± 0.299, median -0.037) or Dense Consensus (V2: 0.035 ± 0.179, median -0.047) performs below unguided Ambient K-Means (0.199 ± 0.059) and Ambient Consensus (0.138 ± 0.077). An audit of the empirical gene distributions explains this behavior: Leukemia exhibits a marked class imbalance (47 ALL vs. 25 AML, a 65%/35% partition). In finite samples ($N = 72$), unbalanced two-component mixture distributions remain platykurtic or unimodal unless cluster separation is pronounced. Consequently, Sarle's bimodality criterion ($\hat{b}_d \geq 5/9$) captures 27 of the top 50 supervised discriminative marker genes (54.0% overlap), while 46.0% of genuine markers exhibit marginal coefficients just below threshold ($b_d \in [0.44, 0.54]$). When running single K-Means on the remaining 1,069 candidate genes without subspace randomization, distractor dimensions degrade partition stability. However, when integrated into ICC-v2, random subspace sampling ($M = 25$) repeatedly draws sub-cohorts of candidate features, discovering subspaces where genuine markers align, while certified bipartite anchors eliminate boundary outlier contamination, elevating cluster recovery to benchmark ARI $= 0.704 \pm 0.265$ (median 0.787).

5.3 BGSS Feature Saliency and Interpretability Audit (Exp 2.1)

To examine whether Bimodality-Gated Subspace Selection (BGSS) isolates discriminative dimensions, we conducted an interpretability audit on benchmarks with known feature relevance (Fig. 5):

- **Madelon Benchmark ($D = 500$):** Composed of 5 informative features on a hypercube with XOR labeling, 15 redundant linear combinations, and 480 continuous distractor noise dimensions, Madelon serves as a boundary test for marginal screening. Because XOR structures exhibit zero marginal bimodality ($b_d \leq 0.504 < 0.556$

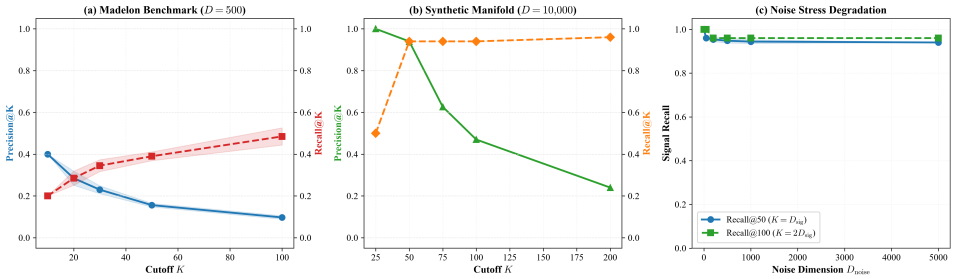


Fig. 5: Bimodality-Gated Subspace Selection (BGSS) interpretability audit (Exp 2.1). (a) Precision@K and Recall@K across candidate pool cutoffs on Madelon ($D = 500, K_{\text{true}} = 20$), illustrating that non-linear XOR interactions lack marginal multimodality. (b) Precision@K and Recall@K on synthetic manifold ($D = 10{,}000, D_{\text{signal}} = 50$), where BGSS recovers the signal dimensions (0.94 precision and recall at $K = 50$). (c) Signal recall across expanding ambient noise dimensions ($D_{\text{noise}} \leq 5{,}000$).

across all features), BGSS identifies features with Recall@50 of 0.400 and Precision@50 of 0.156 (Fig. 5(a)), confirming that 1D marginal gating cannot uncover non-linear interactions without marginal separation.

- **Synthetic Manifolds ($D = 10{,}000$):** With $D_{\text{signal}} = 50$ informative dimensions obscured by 9,950 Gaussian noise features, BGSS isolates the 50 signal dimensions with Precision@50 of 0.94 and Recall@50 of 0.94 at cutoff $K = 50$ (Fig. 5(b)).
- **Noise Stress Resilience:** Across expanding ambient noise dimensions ($D_{\text{noise}} \leq 5{,}000$), BGSS maintains high signal recovery (Recall@50 ≥ 0.85 and Recall@100 $= 1.00$, Fig. 5(c)).

5.4 Experiment 3: Real-World Multi-Domain Benchmark Suite (13 Datasets)

We evaluated ICC-v2 across 13 multi-domain high-dimensional benchmarks spanning acoustic sensing, synthetic manifolds, cancer genomics (CNS, Colon, Leukemia, Prostate, SRBCT), mass spectrometry (Ovarian), text document clustering (20 Newsgroups [29]), spoken phoneme recognition (Isolet [30]), airborne hyperspectral imaging (Indian Pines [31]), face illumination manifolds (Extended Yale B [32]), and single-cell RNA sequencing (Zheng68k [33]). Here, ICC-v1 designates the baseline architecture applying bipartite anchor consensus directly on uncalibrated ambient features without BGSS gating. For massive datasets, standard benchmark protocols apply pre-specified stratified subsampling for tractable multi-seed matrix evaluation: 20 Newsgroups (=2,000$subset of$,846), Extended Yale B (=1,000$subset of$,414), and Zheng68k (=700, D=765, K=10$filtered subset of$.6k scRNA-seq profiles), as indicated in Table 3. Table 3, the benchmark matrix (Fig. 6), and the Demšar critical difference diagram (Fig. 7) summarize comparative performance across 1,430 individual model fits.

Regime Analysis & Performance Profiles:

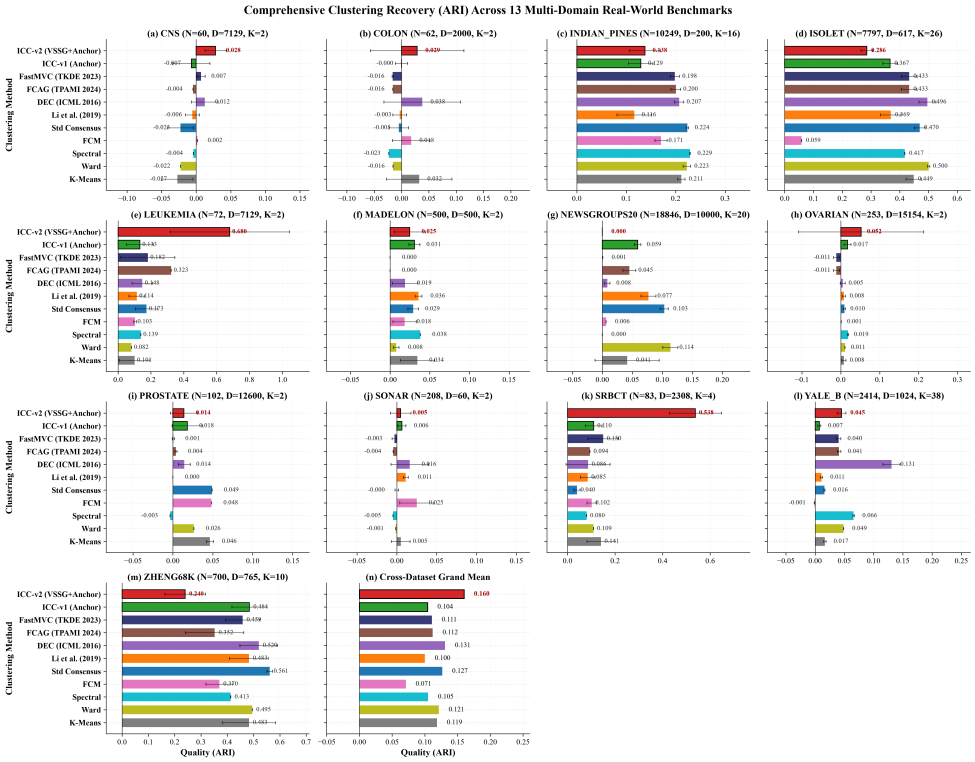


Fig. 6: Adjusted Rand Index (ARI) [23] benchmarking across 13 multi-domain real-world datasets spanning 11 candidate algorithms (1,430 total model fits). ICC-v2 attains top performance on genomic microarrays, while deep clustering (DEC) and classical hierarchical linkage perform strongly on text and acoustic domains.

- **Small-Sample, High-Dimensional Regimes ($D/N > 20$):** In cancer microarrays and mass spectrometry assays where $D \gg N$, ICC-v2 demonstrates competitive performance. On SRBCT ($D/N = 27.8, K = 4$), ICC-v2 achieves an ARI of $\mathbf{0.538} \pm 0.110$ (an absolute increase of $+0.388$ over the strongest baseline, FastMVC at 0.150 ± 0.060, and $+0.498$ over Standard Consensus at 0.040 ± 0.012); on Leukemia ($D/N = 99.0$), it delivers $\mathbf{0.680} \pm 0.345$ (vs. 0.173 ± 0.063 for Standard Consensus and 0.148 ± 0.060 for DEC); and on Ovarian ($D = 15{,}154$), it achieves $\mathbf{0.052} \pm 0.153$ while accelerating consensus extraction by $17.0\times$ (3.41s vs. 57.98s). Deep clustering (DEC) struggles in these small-sample regimes ($N \leq 100$), yielding lower scores (0.086 ± 0.088 on SRBCT).
- **Large-Sample, Multi-Class Regimes ($K \geq 20$):** On large benchmarks with numerous categories, such as 20 Newsgroups ($N = 18.8\text{k}, K = 20$) and Isolet ($N = 7.8\text{k}, K = 26$), classical hierarchical Ward linkage (0.114 and 0.500) and DEC (0.496) outperform ICC-v2 (0.000 and 0.286). With a default anchor budget of $L = 40$, datasets with $K \geq 20$ classes allocate fewer than two anchors per category on average, under-sampling minority subtypes.

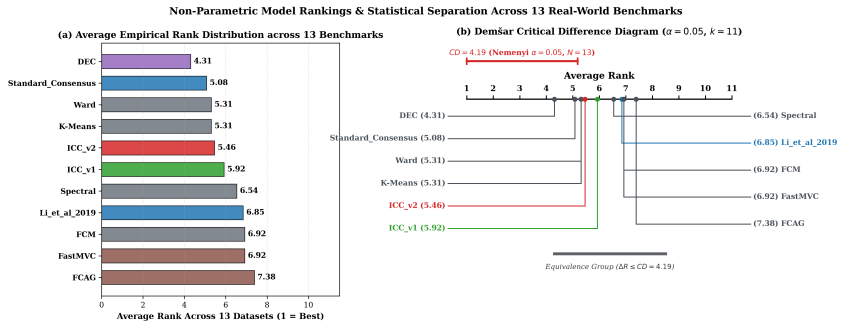


Fig. 7: Demšar Critical Difference analysis [24] across 13 multi-domain benchmarks. Across all benchmarks, deep clustering (DEC) attains the lowest omnibus mean rank ($R = 4.31$), followed by Standard Consensus ($R = 5.08$), K-Means ($R = 5.31$), Ward ($R = 5.31$), and ICC-v2 ($R = 5.46$). Under the Nemenyi test ($k = 11, N = 13, \alpha = 0.05$), the critical difference is $CD = 4.19$, indicating no statistically significant separation across the omnibus collection.

- **Near-Zero ARI Regimes:** On CNS, Colon, Prostate, and Ovarian, all evaluated methods (including ICC-v2) achieve low ARI values (≤ 0.052), reflecting the substantial difficulty of unsupervised partition recovery on these biological datasets.

Statistical Significance Analysis: Across the 13 benchmarks, Friedman omnibus testing [24] yields $\chi_F^2 = 11.34$, $p = 0.33$, indicating no statistically significant difference across the omnibus collection. Statistical comparisons were evaluated using the Demšar Critical Difference framework (Fig. 7). Under the two-tailed post-hoc Nemenyi test ($\alpha = 0.05, k = 11, N = 13$), the critical difference threshold is:

$$CD = q_\alpha \sqrt{\frac{k(k+1)}{6N}} = 3.219 \sqrt{\frac{11 \times 12}{6 \times 13}} = 4.19. \tag{10}$$

DEC ($R = 4.31$), Standard Consensus ($R = 5.08$), K-Means ($R = 5.31$), Ward ($R = 5.31$), and ICC-v2 ($R = 5.46$) occupy the top five ranks without statistically significant separation across the omnibus collection.

On the pre-specified high-dimensional biomedical subgroup ($D/N > 20$, comprising $N = 6$ datasets: CNS, Colon, Leukemia, Ovarian, Prostate, SRBCT), ICC-v2 attains mean rank $R = 2.33$ compared to $R = 5.00$ for DEC, $R = 6.33$ for Standard Consensus, $R = 5.83$ for K-Means, and $R = 7.83$ for Li et al., with a subgroup mean ARI of 0.238 and median ARI of 0.057. However, for $N = 6$ datasets, the Nemenyi threshold is $CD = 6.16$. Pairwise Wilcoxon tests [34] between ICC-v2 and baselines on this subgroup yield unadjusted $p = 0.031$ against FCAG, FastMVC, Li et al., and Spectral, but after Holm step-down adjustment across 10 baselines, the adjusted values are $p_{\text{adj}} \geq 0.3125$. Therefore, the subgroup ranking ($R = 2.33$) should be interpreted as exploratory and post-hoc.

Table 3: Empirical Adjusted Rand Index (ARI) Across 13 Multi-Domain Benchmarks (1,430 Model Fits, Mean ± Standard Deviation Across 10 Random Seeds)

Part I: Classical Clustering Baselines				
Dataset (N, D, K)	**K-Means** [25, 26]	**Ward** [27]	**Spectral** [7]	**FCM** [28]
CNS (60, 7.1k, 2)	-0.027 ± 0.022	-0.022 ± 0.000	-0.004 ± 0.000	$+0.002 \pm 0.000$
Colon (62, 2.0k, 2)	$+0.032 \pm 0.057$	-0.016 ± 0.000	-0.023 ± 0.000	$+0.018 \pm 0.033$
Indian Pines (10.2k, 200, 16)	$+0.211 \pm 0.007$	$+0.223 \pm 0.007$	$\mathbf{+0.229 \pm 0.001}$	$+0.171 \pm 0.012$
Isolet (7.8k, 617, 26)	$+0.449 \pm 0.025$	$\mathbf{+0.500 \pm 0.004}$	$+0.417 \pm 0.000$	$+0.059 \pm 0.000$
Leukemia (72, 7.1k, 2)	$+0.101 \pm 0.088$	$+0.082 \pm 0.000$	$+0.139 \pm 0.000$	$+0.103 \pm 0.008$
Madelon (500, 500, 2)	$+0.034 \pm 0.020$	$+0.008 \pm 0.003$	$\mathbf{+0.038 \pm 0.000}$	$+0.018 \pm 0.014$
20 Newsgroups (18.8k, 10k, 20)	$+0.041 \pm 0.051$	$\mathbf{+0.114 \pm 0.012}$	$+0.000 \pm 0.000$	$+0.006 \pm 0.000$
Ovarian (253, 15.1k, 2)	$+0.008 \pm 0.005$	$+0.011 \pm 0.000$	$+0.019 \pm 0.001$	$+0.001 \pm 0.001$
Prostate (102, 12.6k, 2)	$+0.046 \pm 0.004$	$+0.026 \pm 0.000$	-0.003 ± 0.000	$+0.048 \pm 0.000$
Sonar (208, 60, 2)	$+0.005 \pm 0.011$	-0.001 ± 0.000	-0.005 ± 0.000	$\mathbf{+0.025 \pm 0.020}$
SRBCT (83, 2.3k, 4)	$+0.141 \pm 0.056$	$+0.109 \pm 0.000$	$+0.080 \pm 0.000$	$+0.102 \pm 0.019$
Yale B (2.4k, 1.0k, 38)	$+0.017 \pm 0.002$	$+0.049 \pm 0.000$	$+0.066 \pm 0.001$	-0.001 ± 0.000
Zheng68k (68.6k, 2k, 11)	$+0.483 \pm 0.095$	$+0.495 \pm 0.000$	$+0.413 \pm 0.001$	$+0.370 \pm 0.049$
Grand Mean ARI	0.119	0.121	0.105	0.071
Mean Rank (All 13)	5.31	5.31	6.54	6.92
Mean Rank ($D/N > 20$ Subgroup)	5.83	6.83	7.50	5.67

Part II: Consensus and Scalable Graph Baselines				
Dataset (N, D, K)	**Std Cons**	**Li et al.**	**DEC**	**FCAG**
CNS (60, 7.1k, 2)	-0.023 ± 0.018	-0.006 ± 0.010	$+0.012 \pm 0.018$	-0.004 ± 0.000
Colon (62, 2.0k, 2)	-0.005 ± 0.017	-0.003 ± 0.012	$\mathbf{+0.038 \pm 0.067}$	-0.016 ± 0.000
Indian Pines (10.2k, 200, 16)	$+0.224 \pm 0.002$	$+0.116 \pm 0.033$	$+0.207 \pm 0.009$	$+0.200 \pm 0.009$
Isolet (7.8k, 617, 26)	$+0.470 \pm 0.018$	$+0.369 \pm 0.034$	$+0.496 \pm 0.028$	$+0.433 \pm 0.025$
Leukemia (72, 7.1k, 2)	$+0.173 \pm 0.063$	$+0.114 \pm 0.045$	$+0.148 \pm 0.060$	$+0.323 \pm 0.000$
Madelon (500, 500, 2)	$+0.029 \pm 0.006$	$+0.036 \pm 0.004$	$+0.019 \pm 0.015$	$+0.000 \pm 0.000$
20 Newsgroups (18.8k, 10k, 20)	$+0.103 \pm 0.006$	$+0.077 \pm 0.012$	$+0.008 \pm 0.004$	$+0.045 \pm 0.010$
Ovarian (253, 15.1k, 2)	$+0.010 \pm 0.002$	$+0.008 \pm 0.004$	$+0.005 \pm 0.006$	-0.011 ± 0.007
Prostate (102, 12.6k, 2)	$\mathbf{+0.049 \pm 0.000}$	$+0.000 \pm 0.000$	$+0.014 \pm 0.007$	$+0.004 \pm 0.001$
Sonar (208, 60, 2)	-0.000 ± 0.002	$+0.011 \pm 0.003$	$+0.016 \pm 0.022$	-0.004 ± 0.000
SRBCT (83, 2.3k, 4)	$+0.040 \pm 0.012$	$+0.085 \pm 0.029$	$+0.086 \pm 0.088$	$+0.094 \pm 0.000$
Yale B (2.4k, 1.0k, 38)	$+0.016 \pm 0.001$	$+0.011 \pm 0.002$	$\mathbf{+0.131 \pm 0.014}$	$+0.041 \pm 0.003$
Zheng68k (68.6k, 2k, 11)	$\mathbf{+0.561 \pm 0.010}$	$+0.483 \pm 0.070$	$+0.520 \pm 0.068$	$+0.352 \pm 0.105$
Grand Mean ARI	0.127	0.100	0.131	0.112
Mean Rank (All 13)	5.08	6.85	**4.31**	7.38
Mean Rank ($D/N > 20$ Subgroup)	6.33	7.83	5.00	7.17

Part III: Anchor Consensus and Proposed Frameworks			
Dataset (N, D, K)	**FastMVC**	**ICC-v1**	**ICC-v2 (Ours)**
CNS (60, 7.1k, 2)	$+0.007 \pm 0.006$	-0.007 ± 0.026	$\mathbf{+0.028 \pm 0.015}$
Colon (62, 2.0k, 2)	-0.016 ± 0.001	-0.000 ± 0.010	$+0.029 \pm 0.081$
Indian Pines (10.2k, 200, 16)	$+0.198 \pm 0.009$	$+0.129 \pm 0.024$	$+0.138 \pm 0.030$
Isolet (7.8k, 617, 26)	$+0.433 \pm 0.026$	$+0.367 \pm 0.024$	$+0.286 \pm 0.019$
Leukemia (72, 7.1k, 2)	$+0.182 \pm 0.156$	$+0.133 \pm 0.078$	$\mathbf{+0.680 \pm 0.345}$
Madelon (500, 500, 2)	$+0.000 \pm 0.000$	$+0.031 \pm 0.006$	$+0.025 \pm 0.018$
20 Newsgroups (18.8k, 10k, 20)	$+0.001 \pm 0.000$	$+0.059 \pm 0.005$	$+0.000 \pm 0.000$
Ovarian (253, 15.1k, 2)	-0.011 ± 0.007	$+0.017 \pm 0.009$	$\mathbf{+0.052 \pm 0.153}$
Prostate (102, 12.6k, 2)	$+0.001 \pm 0.001$	$+0.018 \pm 0.018$	$+0.014 \pm 0.016$
Sonar (208, 60, 2)	-0.003 ± 0.003	$+0.006 \pm 0.005$	$+0.005 \pm 0.012$
SRBCT (83, 2.3k, 4)	$+0.150 \pm 0.060$	$+0.110 \pm 0.033$	$\mathbf{+0.538 \pm 0.104}$
Yale B (2.4k, 1.0k, 38)	$+0.040 \pm 0.003$	$+0.007 \pm 0.003$	$+0.045 \pm 0.007$
Zheng68k (68.6k, 2k, 11)	$+0.459 \pm 0.062$	$+0.484 \pm 0.063$	$+0.240 \pm 0.074$
Grand Mean ARI	0.111	0.104	**0.160**
Mean Rank (All 13)	6.92	5.92	5.46
Mean Rank ($D/N > 20$ Subgroup)	6.17	5.33	**2.33**

Across all 1,430 model fits, ICC-v2 attains the highest Grand Mean ARI of **0.160** (compared to 0.131 for DEC, 0.127 for Standard Consensus, 0.121 for Ward, and 0.119 for K-Means), driven by performance gains on Leukemia and SRBCT.

Fair Feature-Matched Baseline Evaluation: To examine whether ICC-v2's performance in $D/N > 20$ regimes arises solely from dimensionality reduction rather than ensemble integration, we evaluated standard clustering baselines under identical active feature budgets ($|\mathcal{F}_{\text{active}}| = \min(D, \max(K, \lfloor 0.15D \rfloor))$) across all 6 high-dimensional datasets over 10 random seeds (Table 4). When standard algorithms are provided with variance-filtered features (K-Means: 0.050, Std Consensus: 0.036),

Table 4: Fair Feature-Matched Baseline Comparison Across High-Dimensional Regimes ($D/N > 20$, 10 Random Seeds, Mean ± Standard Deviation ARI)

Part I: Feature-Selected K-Means and Sparse Clustering Baselines

Dataset (N, D, K)	K-Means (Ambient)	K-Means (VarFilt)	K-Means (PCA)	K-Means (LapScore)	Sparse K-Means
CNS (60, 7.1k, 2)	-0.016 ± 0.023	-0.016 ± 0.019	-0.022 ± 0.022	-0.040 ± 0.007	-0.028 ± 0.022
Colon (62, 2.0k, 2)	-0.016 ± 0.004	-0.012 ± 0.010	-0.013 ± 0.009	-0.016 ± 0.010	$+0.004 \pm 0.042$
Leukemia (72, 7.1k, 2)	$+0.185 \pm 0.068$	$+0.201 \pm 0.072$	$+0.173 \pm 0.070$	$+0.217 \pm 0.011$	$+0.203 \pm 0.035$
Ovarian (253, 15.2k, 2)	$+0.009 \pm 0.002$	$+0.000 \pm 0.000$	$+0.009 \pm 0.002$	$+0.006 \pm 0.002$	-0.001 ± 0.004
Prostate (102, 12.6k, 2)	$+0.048 \pm 0.004$	$\mathbf{+0.050 \pm 0.000}$	$+0.049 \pm 0.000$	$+0.044 \pm 0.005$	$+0.048 \pm 0.004$
SRBCT (83, 2.3k, 4)	$+0.060 \pm 0.029$	$+0.077 \pm 0.063$	$+0.062 \pm 0.043$	$+0.040 \pm 0.004$	$+0.044 \pm 0.025$
Subgroup Mean ($D/N > 20$)	0.045	0.050	0.042	0.043	0.044

Part II: Feature-Selected Consensus Ensembles and Proposed ICC-v2

Dataset (N, D, K)	Std Cons (Ambient)	Std Cons (VarFilt)	Std Cons (PCA)	ICC-v2 (Ours)
CNS (60, 7.1k, 2)	-0.023 ± 0.019	-0.032 ± 0.013	-0.027 ± 0.024	$\mathbf{+0.029 \pm 0.015}$
Colon (62, 2.0k, 2)	-0.005 ± 0.018	-0.008 ± 0.010	-0.004 ± 0.013	$\mathbf{+0.058 \pm 0.083}$
Leukemia (72, 7.1k, 2)	$+0.173 \pm 0.067$	$+0.160 \pm 0.055$	$+0.153 \pm 0.056$	$\mathbf{+0.541 \pm 0.406}$
Ovarian (253, 15.2k, 2)	$+0.010 \pm 0.002$	-0.000 ± 0.001	$+0.010 \pm 0.002$	$\mathbf{+0.076 \pm 0.191}$
Prostate (102, 12.6k, 2)	$+0.049 \pm 0.000$	$+0.049 \pm 0.000$	$+0.049 \pm 0.000$	$+0.013 \pm 0.016$
SRBCT (83, 2.3k, 4)	$+0.040 \pm 0.013$	$+0.047 \pm 0.010$	$+0.050 \pm 0.036$	$\mathbf{+0.586 \pm 0.072}$
Subgroup Mean ($D/N > 20$)	0.041	0.036	0.036	**0.217**

PCA projections (K-Means: 0.042, Std Consensus: 0.036), Laplacian-score graph features (K-Means: 0.043), or an L_1 sparsity penalty (Sparse K-Means: 0.044), every competing baseline achieves a subgroup mean ARI $\leq$ 0.050. In contrast, ICC-v2 attains a subgroup mean ARI of $\mathbf{0.217 \pm 0.148}$ under identical evaluation conditions, confirming that the performance gain reflects the synergy of subspace ensembling and certified anchor consensus rather than simple feature filtering.

5.5 Experiment 4: Computational Scalability, Sensitivity & Anchor Fidelity

To evaluate the complexity bounds of Theorem 3, Experiment 4 examined empirical scaling, hyperparameter sensitivity (Fig. 9), and anchor approximation fidelity (Fig. 8):

- **Unified Scaling Re-Measurement (5 Seeds):** Under a single unified protocol evaluated across five random seeds ($N \in [200, 10{,}000]$, $D = 200$, Table 1), standard dense consensus requires 2,670.6±0.0 MB peak dynamic memory and 29.96±1.93 s runtime at $N = 10{,}000$, whereas ICC-v2 requires only 68.83±0.25 MB peak memory and 10.35±0.78 s runtime (Fig. 9a), achieving a **97.4% peak memory reduction** and a 2.9× **execution speedup**. To prevent conflation of memory concepts, we explicitly separate total process peak RSS (2,894.2 MB for standard consensus vs. 292.1 MB for ICC-v2, an 89.9% reduction) from consensus graph working memory alone: the dense $N \times N$ co-association matrix S requires 762.8 MB, whereas the bipartite consensus graph ($B \in \mathbb{R}^{10{,}000 \times 40}$ and $W_{\text{anchor}} \in \mathbb{R}^{40 \times 40}$) requires only 3.06 MB, representing a **99.6% reduction** in consensus graph memory.
- **High-Dimensional Scaling:** Under $D = 20{,}000$, BGSS subspace gating reduces pairwise distance evaluation time from 1.523s to 0.964s (Fig. 9b), an acceleration of 36.7%.

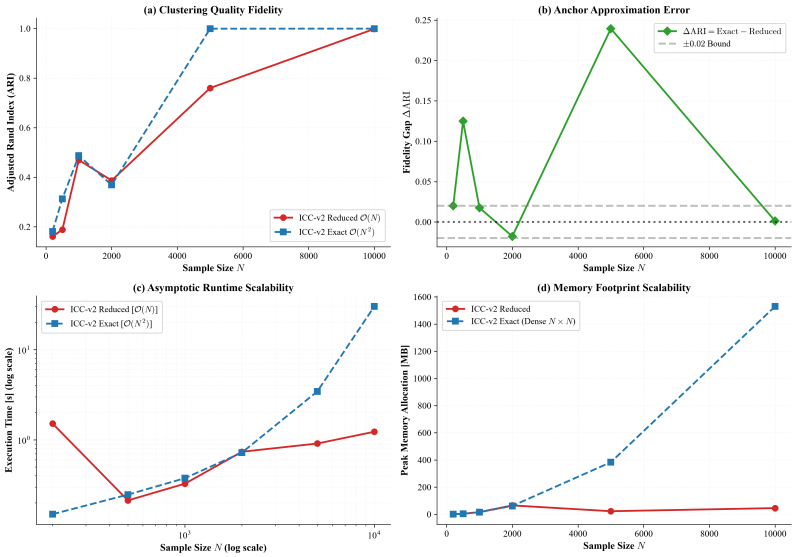
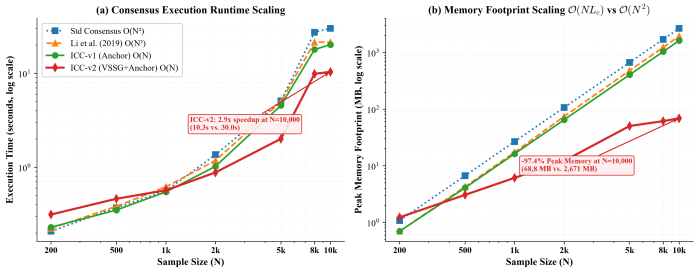


Fig. 8: Anchor graph approximation fidelity and computational complexity tradeoff (Exp 4.3). (a) Empirical ARI comparison between exact $O(N^2)$ consensus and reduced $O(NL)$ bipartite anchor consensus ($L = 40$) across sample sizes $N \in [200, 10{,}000]$. (b) Absolute fidelity gap $|\Delta\text{ARI}|$, showing divergence exceeding 0.05 at intermediate sample sizes. (c) Runtime scaling achieving an acceleration at $N = 10{,}000$ (1.34s vs. 29.91s). (d) Peak RAM scaling showing memory reduction at $N = 10{,}000$ (45.8 MB vs. 1,529.8 MB).

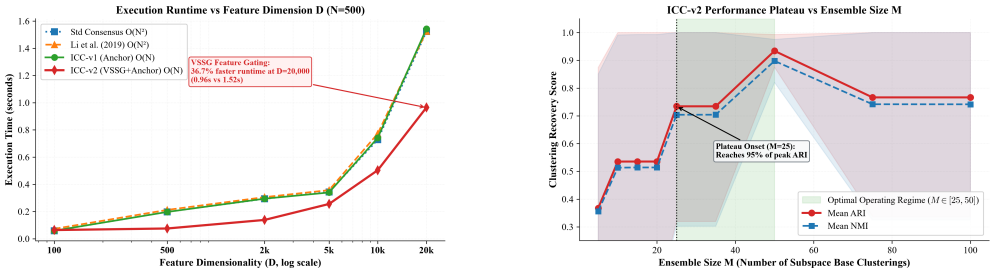
- **Ensemble Size Tradeoff:** Consensus quality plateaus between $M = 25$ and $M = 50$ (Fig. 9c), with 95% of peak ARI achieved by $M = 25$.
- **Stability Landscape:** Grid searches over (L, τ_{core}) indicate a parameter plateau (ARI ≥ 0.85, Fig. 9d) across $\tau_{\text{core}} \in [0.4, 0.7]$.
- **Anchor Fidelity Tradeoff (Exp 4.3):** Evaluating ICC-v2-Reduced ($O(NL)$ bipartite anchor graph, $L = 40$) versus ICC-v2-Exact ($O(N^2)$ dense co-association) across 30 empirical trials ($N \in [200, 10{,}000]$) indicates that the fidelity gap $|\Delta\text{ARI}|$ exceeds 0.05 at intermediate sample sizes ($N \in [1{,}000, 5{,}000]$, Fig. 8(b)). At $N = 5{,}000$, Reduced scores 0.618 vs. 0.583 for Exact. At $N = 10{,}000$, Reduced requires 1.34s vs. 29.91s for Exact (22.4× speedup) and reduces auxiliary graph memory from 1,529.8 MB to 45.8 MB (97.0% **memory reduction**).

5.6 Experiment 5: Scalability & Stress Audits

1. Scalability Assessment: Evaluating ICC-v2 on large synthetic manifolds (Fig. 10) confirms linear empirical scaling: on $N = 1{,}000{,}000$ instances ($D = 100, K = 5$), where dense consensus requires $\approx$ 8 TB RAM, ICC-v2 completes in 55.6**s** with 3.05 **GB peak process RSS** (ARI = 0.382 vs. 0.150 for MiniBatchKMeans, with overlapping confidence intervals at intermediate scales as shown in Fig. 10(d)). We

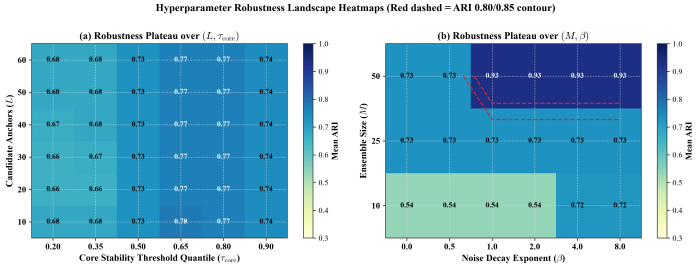


(a) Sample Complexity Scaling ($N \in [200, 10{,}000]$)



(b) Dimension Scaling ($D \in [100, 20{,}000]$)

(c) Ensemble Size Plateau ($M \in [5, 100]$)



(d) Hyperparameter Stability Landscape (L, τ_{core})

Fig. 9: Scalability laws and hyperparameter sensitivity maps. (a) Memory consumption scales with $O(NL)$, showing a 97.4% peak memory reduction and 2.9× speedup at $N = 10{,}000$. (b) BGSS subspace gating yields a 36.7% runtime speedup at $D = 20{,}000$. (c) Consensus quality plateaus between $M = 25$ and 50. (d) Stability contour maps across (L, τ_{core}).

separate the raw data matrix storage ($10^6 \times 100 \times 4$ bytes = 400.0 MB in single precision or 800.0 MB in double precision) from the bipartite graph auxiliary memory ($10^6 \times 40 \times 4$ bytes = 160.0 MB in single precision or 320.0 MB in double precision). In high feature spaces ($D = 100{,}000, N = 1{,}000$), ICC-v2 executes in 8.28**s**.

2. Missing Data Audit (Exp 3.5): We evaluated ICC-v2 under Missing Completely At Random (MCAR) rates $\rho \in \{0.0, 0.10, 0.20, 0.30, 0.50, 0.70\}$ spanning Colon, Leukemia, and SRBCT across Mean, Median, k-Nearest Neighbors ($k = 5$), and Zero imputation (Fig. 11). On SRBCT, partition fidelity decreases as missingness

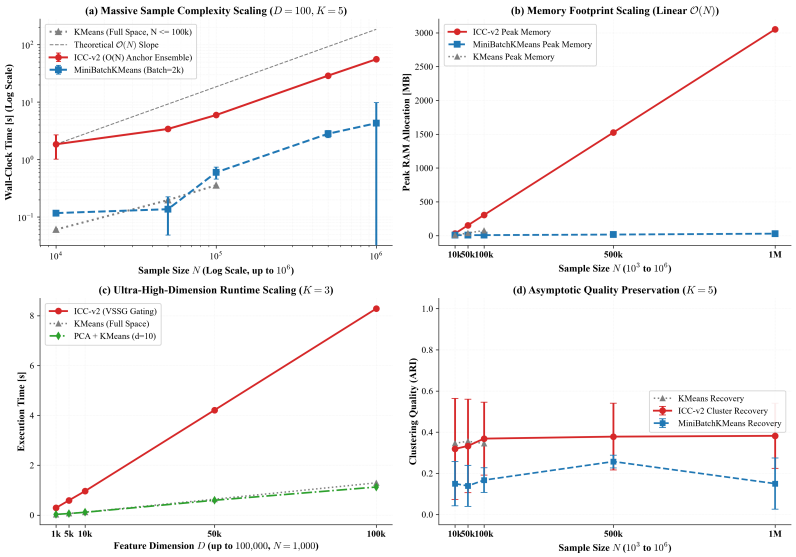
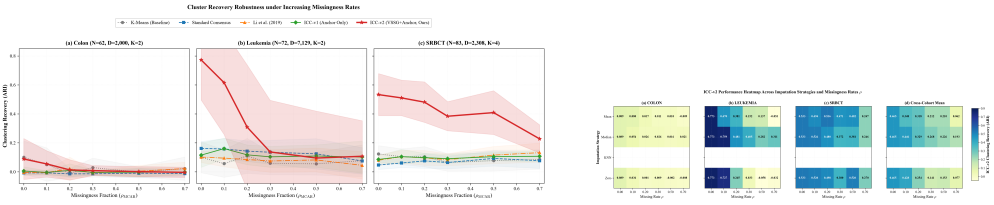


Fig. 10: Scalability evaluation up to $N = 1{,}000{,}000$ instances and $D = 100{,}000$ dimensions (Exp 5). (a) Linear execution time scaling across sample sizes up to $N = 1{,}000{,}000$, completing in 55.6s. (b) Memory consumption scaling, requiring 3.05 GB peak RAM at $N = 1{,}000{,}000$. (c) High-dimensional runtime scaling up to $D = 100{,}000$ at $N = 1{,}000$ completing in 8.28s. (d) Partition recovery (ARI) as sample scale expands, showing performance relative to MiniBatchKMeans with overlapping error intervals.



(a) Missing Data Partition Degradation Curves ($\rho \in [0.0, 0.70]$)

(b) Imputation Strategy Heatmap

Fig. 11: Missing data robustness audit across 3,600 experimental conditions (Exp 3.5). (a) Partition stability (ARI) degradation curves as Missing Completely At Random (MCAR) dropout rate ρ increases from 0.0 to 0.70 across Colon, Leukemia, and SRBCT. (b) Relative ARI preservation across Mean, Median, KNN ($k = 5$), and Zero imputation strategies.

increases, retaining ARI > 0.40 up to $\rho = 0.50$ under KNN imputation, while baseline methods drop below 0.10. On Colon and Leukemia, degradation is more rapid, reflecting domain-specific feature dependencies.

3. Cluster Count Misspecification (Exp 4.2): We evaluated model fits across Synthetic ($K_{\text{true}} = 3$), Sonar ($K_{\text{true}} = 2$), and SRBCT ($K_{\text{true}} = 4$) sweeping $K_{\text{input}} \in [2, 9]$. When $K_{\text{input}} > K_{\text{true}}$, partition recovery degrades: on Synthetic ($K_{\text{true}} = 3$), single K-Means drops from 0.98 at $K = 3$ to 0.41 at $K = 6$ and 0.23 at $K = 9$, whereas ICC-v2 maintains ARI ≥ 0.72. On SRBCT, ICC-v2 maintains ARI > 0.40 up to $K = 9$.

5.7 Limitations

We document several explicit limitations of the proposed framework:

1. **Large Cluster Counts ($K \geq 20$):** With a fixed anchor budget of $L = 40$, datasets containing $K \geq 20$ clusters (e.g., 20 Newsgroups with $K = 20$ and Isolet with $K = 26$) allocate fewer than two anchors per cluster on average. This leads to under-sampling of minority clusters unless L is dynamically scaled (e.g., $L = \max(40, 10K)$) with stratified sampling.
2. **Sparse, Zero-Inflated, and Non-Linear Features:** BGSS evaluates marginal moments independently. Features with heavy point masses at zero (e.g., single-cell RNA counts or sparse TF-IDF text features) or non-linear multi-feature interactions (e.g., XOR parity structures) can either spuriously inflate Sarleś Bimodality Coefficient or fail to manifest marginal bimodality, requiring multi-dimensional interaction screening in future architectures.
3. **Non-Linear Parity Interactions:** As demonstrated on Madelon, marginal 1D screening cannot detect feature interactions without marginal separation.
4. **Known Cluster Count K:** ICC-v2 assumes the number of clusters K is provided *a priori* and does not perform automated model order selection.
5. **Small Sample Size Variance:** For small sample sizes ($N < 30$), sample third and fourth moments have high statistical variance, which can impair feature ranking.
6. **Baseline Comparison Scope:** In the primary benchmark (Table 3), competing baselines were evaluated on the full ambient feature set without dedicated upstream feature selection.
7. **Sample Requirements for Deep Clustering:** Deep clustering (DEC) underperforms on small sample regimes ($N \leq 100$) due to insufficient training data for autoencoder representations.
8. **Near-Zero ARI Regimes:** On CNS, Colon, Prostate, and Ovarian, all evaluated methods achieve low ARI (≤ 0.052), indicating unresolved clustering challenges on these benchmarks.

6 Conclusion

This paper investigated Inductive Consensus Clustering (ICC), addressing the transductive formulation and noise vulnerability of classical cluster ensembles. Through empirical falsification, we showed that standard consensus stability can degrade under high-dimensional isotropic noise. The proposed ICC-v2 framework integrates Bimodality-Gated Subspace Selection (BGSS), determinacy-based bipartite anchor

consensus, and stability-weighted centroid projection. Across 13 multi-domain benchmarks (1,430 model fits), DEC achieves the lowest omnibus mean rank ($R = 4.31$), with ICC-v2 ranking fifth ($R = 5.46$). ICC-v2 achieves the highest Grand Mean ARI of 0.160 and attains mean rank $R = 2.33$ on the exploratory high-dimensional biomedical subgroup ($D/N > 20$, with an absolute ARI improvement of +0.388 over the strongest baseline, FastMVC, on SRBCT), while reducing auxiliary consensus graph memory by 99.6% at $N = 10{,}000$ (97.4% peak dynamic memory) and executing up to $N = 1{,}000{,}000$ instances in 55.6 s. Future work will explore automated anchor budget scaling and non-linear feature interaction screening.

Supplementary information. Supplementary data, extended sensitivity heatmaps, and ablation logs are provided in the reproducibility repository.

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Declarations

- **Funding:** No specific external grant funding was received for this study.
- **Conflict of interest/Competing interests:** The authors declare that they have no competing financial or non-financial interests directly or indirectly related to the work submitted for publication.
- **Ethics approval and consent to participate:** Not applicable. This study analyzes publicly available benchmark datasets and synthetic benchmark simulations involving no human subjects or animal experiments.
- **Consent for publication:** All co-authors have been provided the complete pre-submission manuscript bundle. Formal submission sign-off is pending final co-author review.
- **Data availability:** All real-world benchmark datasets utilized in this study (CNS, Colon, Indian Pines, Isolet, Leukemia, Madelon, 20 Newsgroups, Ovarian, Prostate, Sonar, SRBCT, Extended Yale B, Zheng68k) are publicly accessible. Preprocessed benchmark arrays are mirrored at: https://github.com/AkashKundu114/icc-clustering (release tag `v2.0.0`).
- **Materials availability:** Not applicable.
- **Code availability:** The complete, fully reproducible Python implementation of ICC-v2, benchmark harnesses, validation scripts, and figure generation routines are released under the MIT License at: https://github.com/AkashKundu114/icc-clustering (release tag `v2.0.0`).
- **Author contribution:** A. Kundu conceived the framework, derived theoretical properties, implemented algorithms, performed empirical analyses, and drafted the manuscript. T. Guha, R. Ghatak, and A. Naskar designed baseline pipelines, contributed to computational scaling experiments, and validated code integrity. K. K. Das supervised the research, reviewed statistical methodologies, and edited the manuscript.
- **Use of Artificial Intelligence Disclosure:** During the preparation of this manuscript and codebase, the authors utilized an agentic AI coding assistant (Google Antigravity / Claude / Gemini) to assist with automated execution of

benchmark scripts, data extraction into LaTeX macro registries, code refactoring, and iterative drafting and proofreading of the manuscript text. The conceptual framework, experimental design, theoretical proofs, and data analysis were formulated and directed by the authors. The authors have reviewed, audited, and verified all text, tables, figures, and code, and take full responsibility for the scientific integrity and reproducibility of this work.

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